Case 1: Equity in Engineering and Computing Practices

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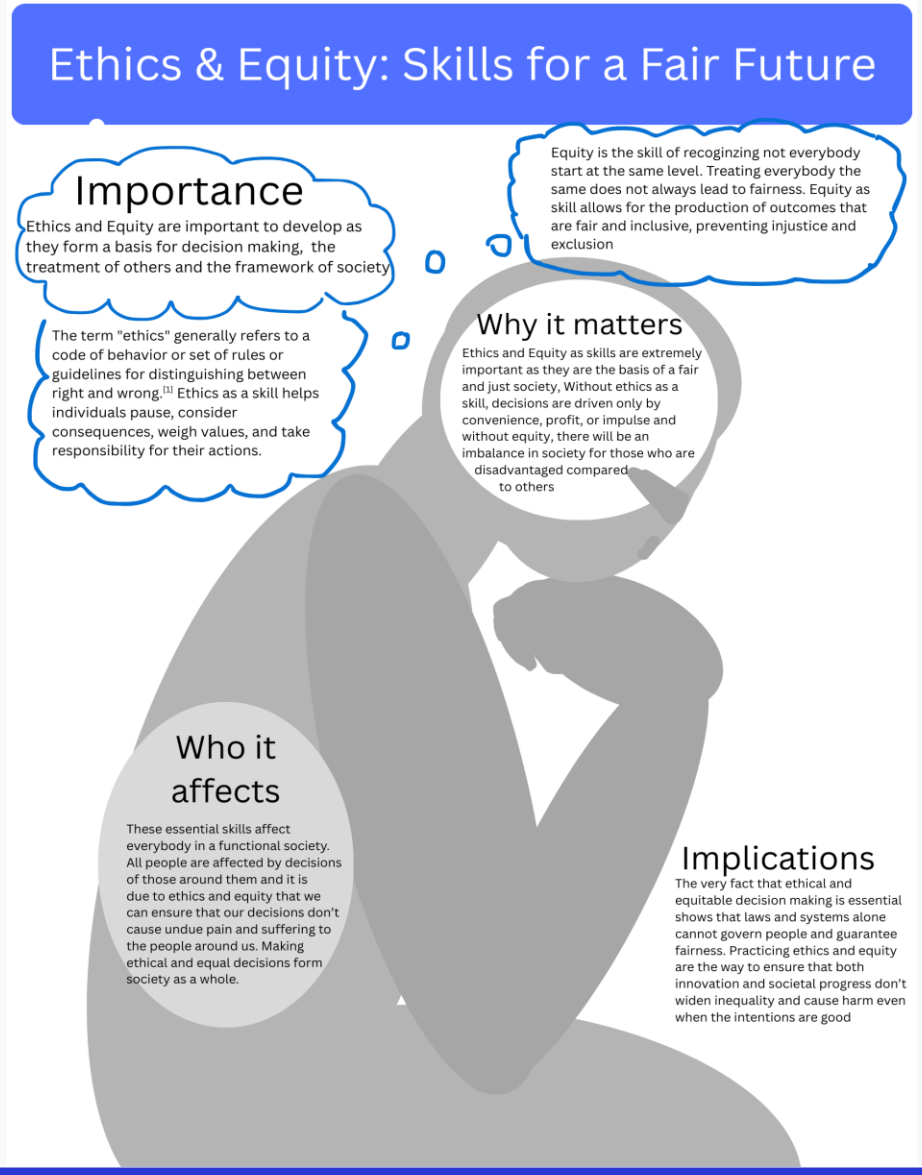


References:[1] R. Pool, "*D. Ethics, Equity, and Responsibility*" https://www.nagb.gov/naep-subject-areas/technology-and-engineering-literacy/framework-archive/2014-technology-framework/toc/ch_2/society/society4.html (accessed Jan 29, 2026).

Second Slide: Cultivating Equity in Education

"We don't see things as they are. We see them as we are." — Anaïs Nin[1]

What the quote means in education?

- In engineering and computing education, teachers and institutions often believe their systems are neutral and fair.
- However, this quote reminds us that **our background, experiences, and assumptions shape how we judge students**.
- As a result, what seems "fair" to one group may unintentionally disadvantage others.

Debate: Is standardization truly fair?

- One view:
  Standardized exams, fixed grading rubrics, and uniform teaching methods ensure fairness and objectivity.
- Opposing view:
  These approaches often favor students who already match the dominant academic culture, language, or prior exposure to computing.

Our Position

- Inequity in engineering and computing education is often not intentional but emerges from unexamined perspectives held by instructors and institutions. Anaïs Nin's statement highlights that perception shapes practice.[2]

References:[1] A. Nin, Seduction of the Minotaur, Chicago, IL, USA: Swallow Press, 1961.
[2] D. Riley, "Engineering and Social Justice," Synthesis Lectures on Engineers, Technology, and Society, vol. 3, no. 1, pp. 1–152, 2008.

Third Slide: Equity in Design & Development

Why design and development decisions matter

- Equity in computing is not only a social issue; it is shaped by **technical choices** made during design and development.
- Decisions about data, interfaces, and testing directly affect **who can access a system and who may be harmed by it**.

Issue:
Models trained on unbalanced datasets may produce biased outcomes for certain groups.

Counterpoint: Developers may argue that using larger datasets improves accuracy.

Algorithmic bias

Trade-off & mitigation:
Accuracy alone is insufficient; teams should audit training data and evaluate models across demographic subgroups [1].

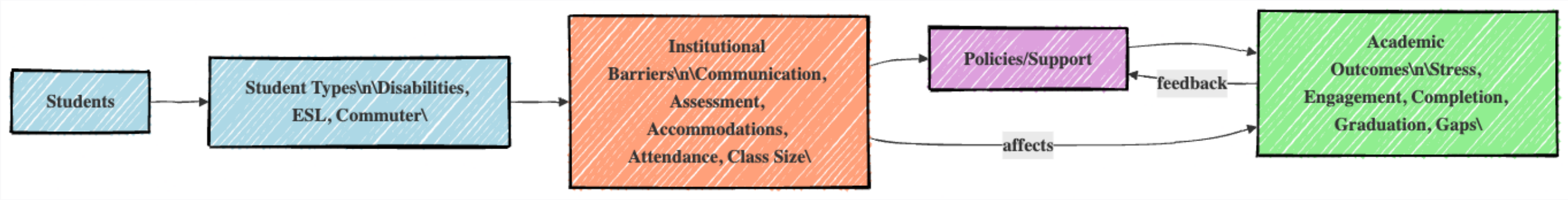
Issue:
Models trained on unbalanced datasets may produce biased outcomes for certain groups.

Connections:
Following accessibility standards early (e.g., WCAG) reduces long-term redesign costs and improves usability for all users [2].

Counterpoint:
Developers may argue that using larger datasets improves accuracy.

References:[1] S. Barocas, M. Hardt, and A. Narayanan, Fairness and Machine Learning, Cambridge, MA, USA: fairmlbook.org, 2019.
[2] World Wide Web Consortium (W3C), “Web Content Accessibility Guidelines (WCAG) 2.1,” 2018. [Online]. Available: https://www.w3.org/TR/WCAG21/

Fourth Slide: Equity in Faculty & Student Academic Experiences



- **Student experiences:** Students with disabilities, first-generation, international, or working students often face *invisible barriers* such as rigid deadlines, inaccessible exams, and unclear expectations [1].

- **Faculty experiences:** Instructors aim for fairness but face **constraints** like large class sizes, limited training, and time pressure [1].

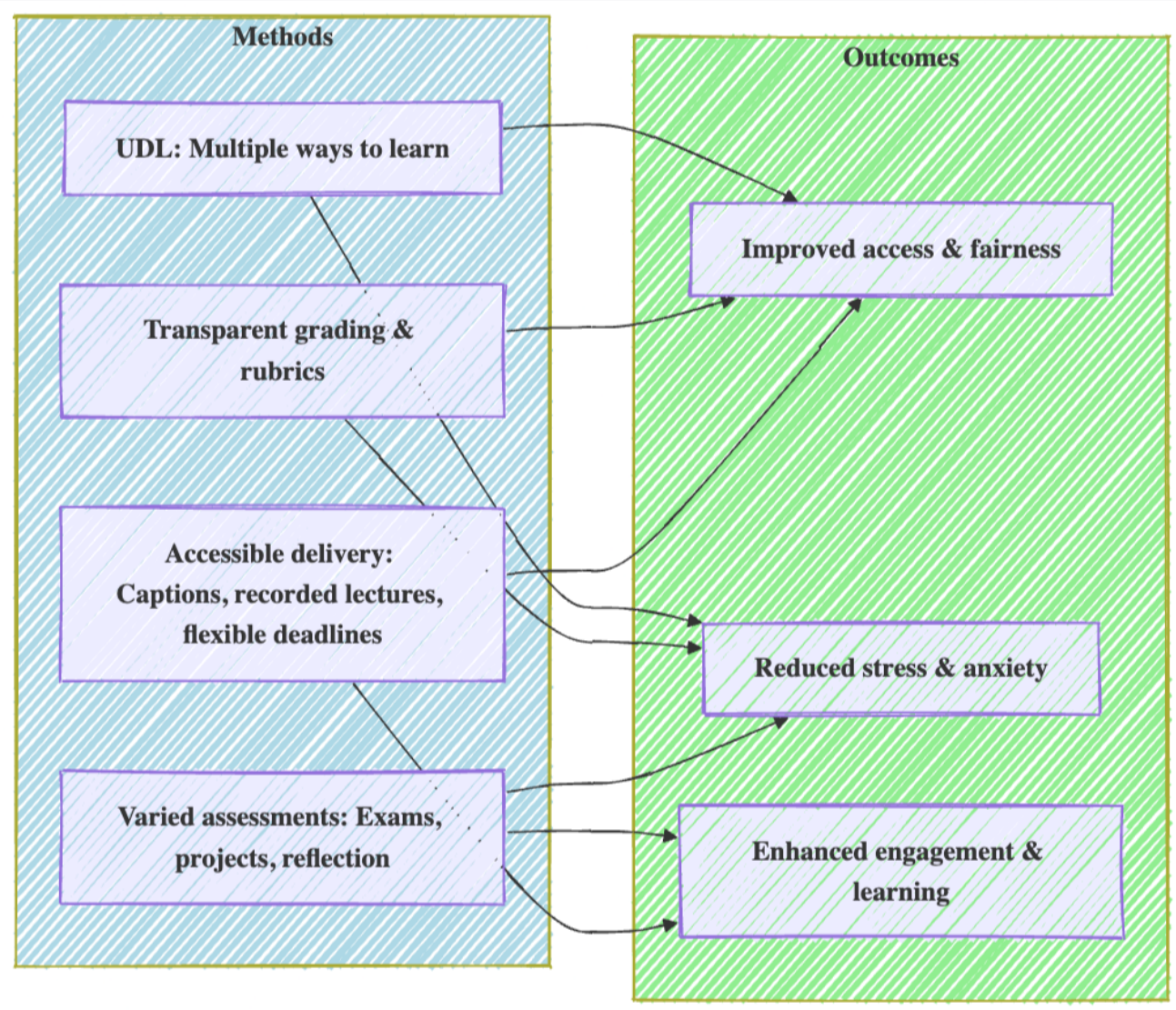
- **Fairness issue:** Treating everyone the same does not guarantee equal opportunity or outcomes [2].

- **Counterpoint & response:**
  - *Counterpoint:* Accommodations can feel unfair or difficult to manage.
  - *Response:* Proactive, structured equity policies reduce last-minute exceptions and improve consistency [1], [2].

References: [1] E. A. Lombardi, R. Murray, and M. Gerdes, “College faculty and inclusive instruction,” *Journal of*

Inclusive methods (specific):

- **Universal Design for Learning (UDL):** Multiple ways to access content and demonstrate learning [3].
- **Transparent grading & rubrics:** Clear expectations reduce bias and ambiguity [4].
- **Varied assessments:** Exams, projects, participation, and reflection allow students to show strengths in different ways [3].
- **Accessible delivery:** Captions, recorded lectures, and structured flexibility improve learning for all [3], [4].



Why it works (debate-style, with citations):

- **Benefits:** Improves access, reduces stress, and enhances learning outcomes [3].
- **Limitations:** Increased preparation time, grading complexity, scalability concerns [4].
- **Mitigation:** Shared rubrics, reusable materials, and institutional support tools [4].

References: [3] CAST, *Universal Design for Learning Guidelines*, Version 2.2, 2018.